Problem 5.16.2. The **content** $c(\lambda)$ of a Young diagram λ is the sum $\sum_j \sum_{i=1}^{\lambda_j}(i-j)$. Let $C = \sum_{i<j}(ij) \in \mathbb{C}[S_n]$ be the sum of all transpositions. Show that C acts on the Specht module V_λ by multiplication by $c(\lambda)$.

Problem 5.16.3. (a) Let V be any finite dimensional representation of S_n. Show that the element $E := (12) + \cdots + (1n)$ is diagonalizable and has integer eigenvalues on V which are between $1-n$ and $n-1$.

Hint: Represent E as $C_n - C_{n-1}$, where $C_n = C$ is the element from Problem 5.16.2.

(b) Show that the element $(12) + \cdots + (1n)$ acts on V_λ by a scalar if and only if λ is a rectangular Young diagram, and compute this scalar.

5.17. The hook length formula

Let us use the Frobenius character formula to compute the dimension of V_λ. According to the character formula, $\dim V_\lambda$ is the coefficient of $x^{\lambda+\rho}$ in $\Delta(x)(x_1 + \cdots + x_N)^n$. Let $l_j = \lambda_j + N - j$. Then, using the determinant formula for $\Delta(x)$ and expanding the determinant as a sum over permutations, we get

$$\begin{aligned}\dim V_\lambda &= \sum_{s\in S_N: l_j \geq N-s(j)} (-1)^s \frac{n!}{\prod_j (l_j - N + s(j))!} \\ &= \frac{n!}{l_1! \dots l_N!} \sum_{s\in S_N} (-1)^s \prod_j l_j(l_j-1)\dots(l_j - N + s(j) + 1) \\ &= \frac{n!}{\prod_j l_j!} \det(l_j(l_j-1)\dots(l_j - N + i + 1)).\end{aligned}$$

Using column reduction and the Vandermonde determinant formula, we see from this expression that

$$\dim V_\lambda = \frac{n!}{\prod_j l_j!} \det(l_j^{N-i}) = \frac{n!}{\prod_j l_j!} \prod_{1\leq i<j\leq N} (l_i - l_j) \tag{5.17.1}$$

(where $N \geq p$).

In this formula, there are many cancellations. After making some of these cancellations, we obtain the hook length formula. Namely, for a square (i,j) in a Young diagram λ $(i,j \geq 1, i \leq \lambda_j)$, define the